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Tutorial Sheet
Fibonacci Heap

Q1) What is the amortized time complexity of the extract-min operation in a Fibonacci heap?

a) O(log n)
b) O(1)
c) O(n)
d) O(n log n)

Q2) What is the worst-case time complexity of merging (unioning) two Fibonacci heaps?

a) O(log n)
b) O(1)
c) O(n)
d) O(n log n)

Q3) Why do we mark nodes in a Fibonacci heap during a decrease-key operation?

a) To identify which node contains the absolute minimum value in the heap.
b) To mark nodes that need to be permanently deleted from memory.
c) To keep track of parents that have lost a child, ensuring sufficient potential for amortized analysis.
d) To prevent the root list from growing too large.

Q4) What is the maximum number of root nodes possible in an n-node Fibonacci heap (i) at any point in time and (ii) immediately after Extract-Min?

a) O(log n), O(log n)
b) n, O(log n)
c) n, n
d) O(log n), n

Q5) What is wrong with the following code of insertion in Fibonacci heap.

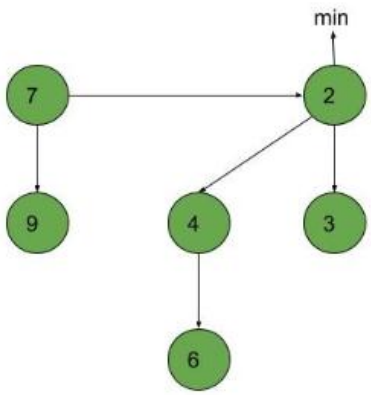
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FIB-INSERT(H, x)
degree[x]= 0
p[x]=  NIL
child[x] =NIL
left[x] =x
right[x] =x
mark[x] =FALSE
concatenate the root list containing x with root list H
if min[H] = NIL or key[x] > key[min[H]]
then min[H]= x
n[H]= n[H] + 1
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a) 2
b) 4 and 5
c) 6
d) 9

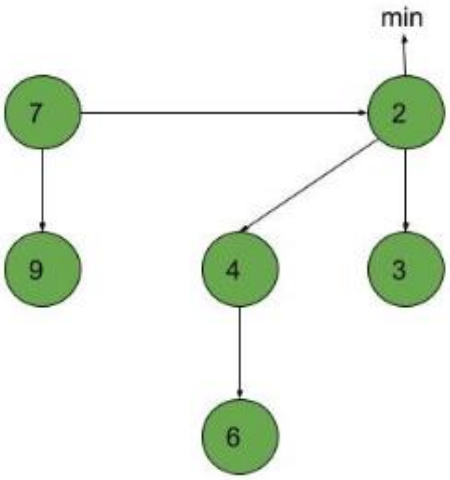
Q6) Choose the option with function having same complexity for a Fibonacci heap.

a) Insertion, Union
b) Insertion, Deletion
c) extract_min, insertion
d) Union, delete

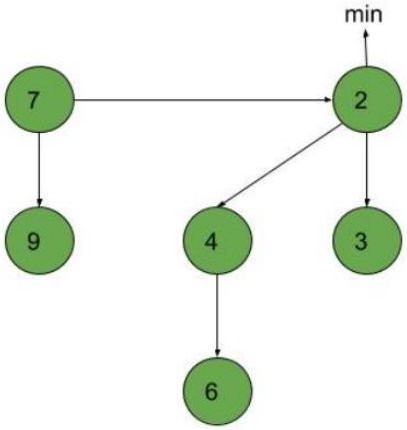
Q7) Apply Extract min operation to following Fibonacci heap.



Q8) Decrease the value of node with key 9 to key 1 in the given Fibonacci heap.

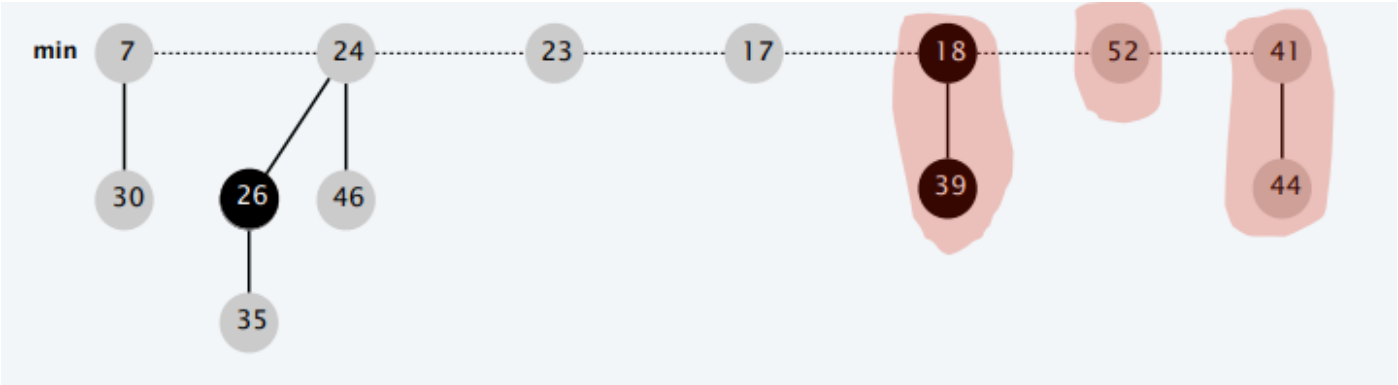


Q9) Delete the node with key value 6 in the given Fibonacci heap.



Delete 6

Q10) Apply consolidation step to this Fibonacci heap such that no two trees should have same degree.



Reference: Kaplan, H. and Tarjan, R.E., 1999. *New heap data structures*. Technical Report TR-597-99, Department of Computer Science, Princeton University.